

AUTONOMOUS RC CAR WEEK 1 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

THE OVERARCHING GOAL FOR THIS TECHNOLOGY

Navigate through deep forests to make labor like forestry and mining safer for humans by using AI to map out unknown and dangerous terrain.



Research

- Understand RC car workings
- Learn how autonomous vehicles function
- Research ways to replace the radio signal communication system, to instead use the Arduino
- Find ways to use cameras/sensors.

Obstacle Avoidance

- Obstacle avoidance uses cameras/sensors to detect when near something
- Some obstacles could be trees or animals
- Also map out obstacles eventually

Code/Software

- We start with obstacle detection
- From there focus on how to move away from obstacles
- We need to figure out what software to use that could make this possible, along with continuous troubleshooting

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

Step 04 (Bonus)

Can it create paths based on given targets?

Example/ Placeholder Goals

Milestone 1

Upload code to the robot using the arduino, for it to move forward to a certain point, turn 180 degrees one way, turn 180 degrees the other way, and then move backwards back to the start.

Milestone 2

Have the robot drive in Stevens park, and stop every time it encounters an obstacle it can't just go through, 10 obstacles in a row without errors.

Milestone 3

Drive around in Stevens Park for 2 minutes without hitting any obstacles.

Milestone 4

Reach a target given to it in Stevens Park within 2 minutes, while still avoiding obstacles.

Hardware Needed

Initial Purchases (~\$50):

- ESP32 microcontroller ("brain" of system)
- Buck converter (high voltage -> low voltage)
- Jumper wires
- ESP32 camera

Future Purchases (~\$60):

- GPS module
 - Set home point
 - Log routes
- IMU (Inertial Measurement Unit)
 - Measures acceleration, direction, and pitch angles of car

**~\$130 Total
Anticipated Budget**

Preliminary Research Completed

- Explored necessary hardware and software moving forward
- ESC (Electronic Speed Controller)
 - Controls RC motor
 - Can be connected to external receiver/controller
- Established development roadmap and milestones



Feedback

- We learned to research Roomba technology because they were successful in solving a similar issue
- Need to get a shopping list to the professor ASAP so we can continue forward
- Have ambitious goals by the end of the summer, but also have good milestones to complete them

THANK YOU

Any questions?